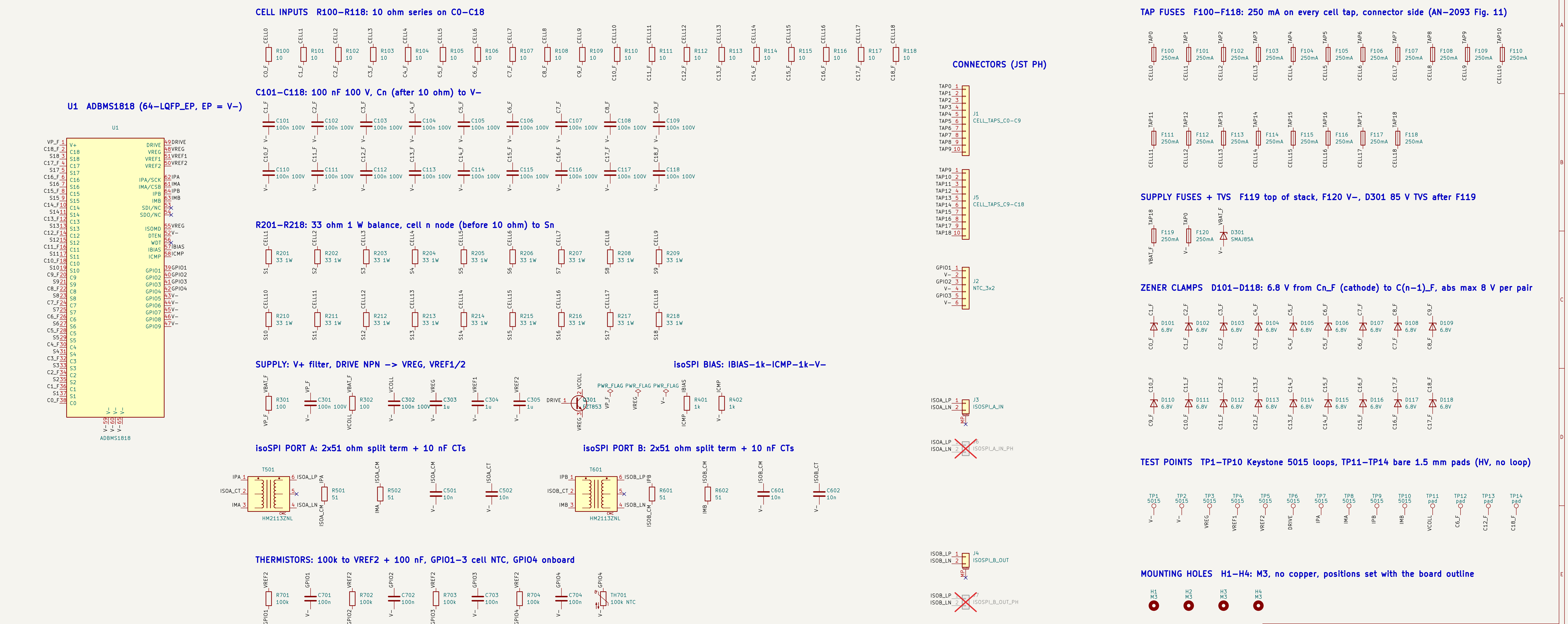
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.


CELL INPUTS R100–R118: 10 ohm series on C0–C18
R100–R118 10
CELL0–CELL18
C0_F–C18_F
C101–C118: 100 nF 100 V, Cn (after 10 ohm) to V−
C101–C118 100n 100V
U1 ADBMS1818 (64-LQFP_EP, EP = V−)
U1
ADBMS1818
V+
DRIVE
VREG
VREF1
VREF2
IPA/SCK
IMA/CSB
IPB
IMB
SDI/NC
SDO/NC
ISOMD
DTEN
WDT
IBIAS
ICMP
GPIO1–GPIO9
C0–C18
S1–S18
R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn
R201–R218 33 1W
SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2
R301 100
C301 100n 100V
R302 100
C302 100n 100V
C303 1u
C304 1u
C305 1u
Q301 FZT853
PWR_FLAG
isoSPI BIAS: IBIAS−1k−ICMP−1k−V−
R401 1k
R402 1k
isoSPI PORT A: 2x51 ohm split term + 10 nF CTs
T501 HM2113ZNL
CMC
R501 51
R502 51
C501 10n
C502 10n
isoSPI PORT B: 2x51 ohm split term + 10 nF CTs
T601 HM2113ZNL
CMC
R601 51
R602 51
C601 10n
C602 10n
THERMISTORS: 100k to VREF2 + 100 nF, GPIO1−3 cell NTC, GPIO4 onboard
R701 100k
C701 100n
R702 100k
C702 100n
R703 100k
C703 100n
R704 100k
C704 100n
TH701 100k NTC
CONNECTORS (JST PH)
J1 CELL_TAPS_C0–C9
J5 CELL_TAPS_C9–C18
J2 NTC_3x2
J3 ISOSPI_A_IN
ISOSPI_A_IN_PH
J4 ISOSPI_B_OUT
ISOSPI_B_OUT_PH
TAP FUSES F100–F118: 250 mA on every cell tap, connector side (AN-2093 Fig. 11)
F100–F118 250mA
TAP0–TAP18
SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119
F119 250mA
F120 250mA
D301 SMAJ85A
ZENER CLAMPS D101–D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair
D101–D118 6.8V
TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)
TP1–TP10 5015
TP11–TP14 pad
MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline
H1–H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User
Date: 2026−11−06
Rev: 0−draft
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